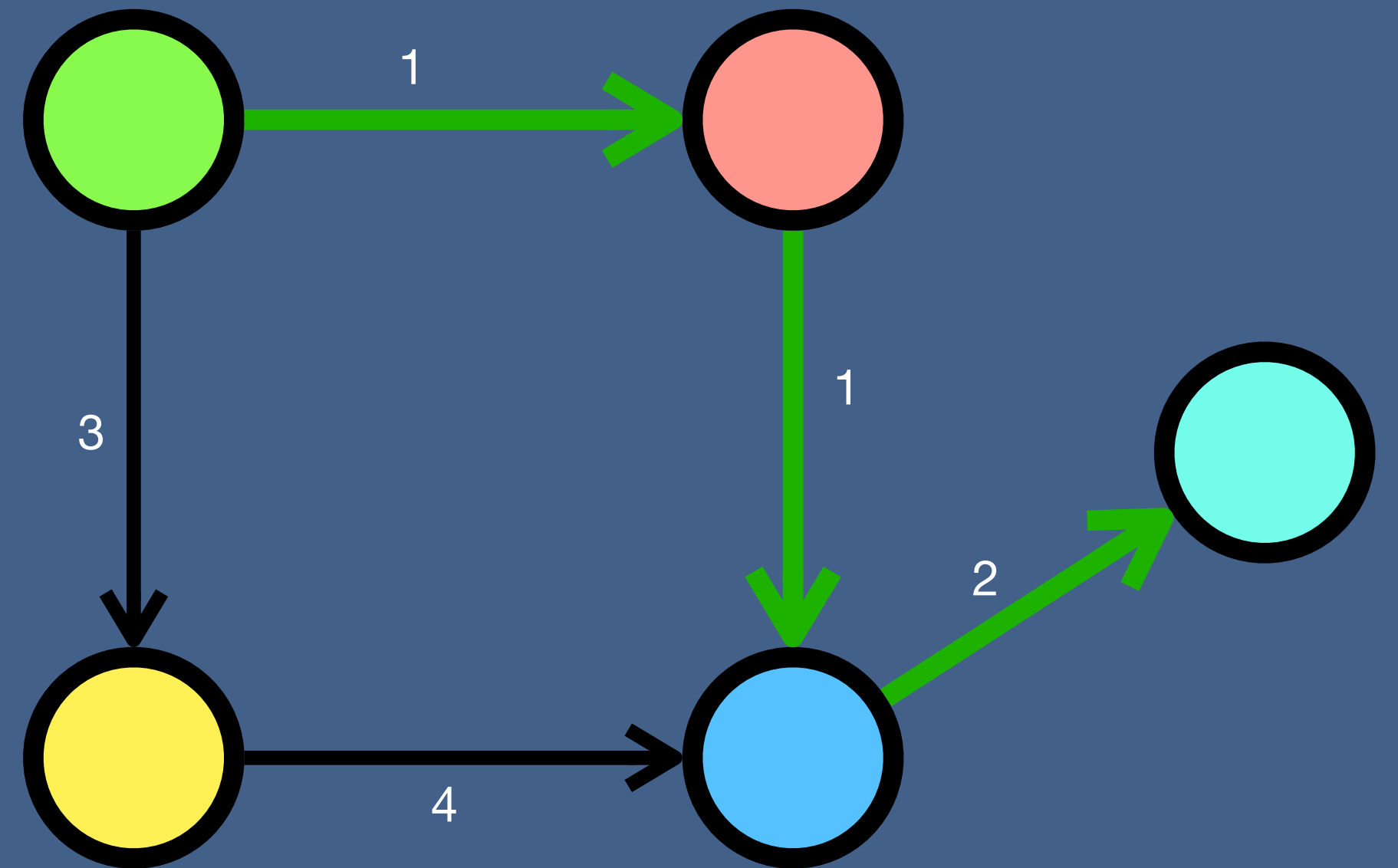
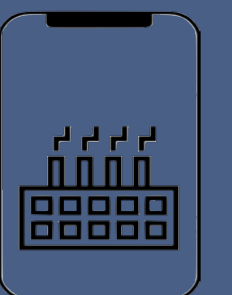
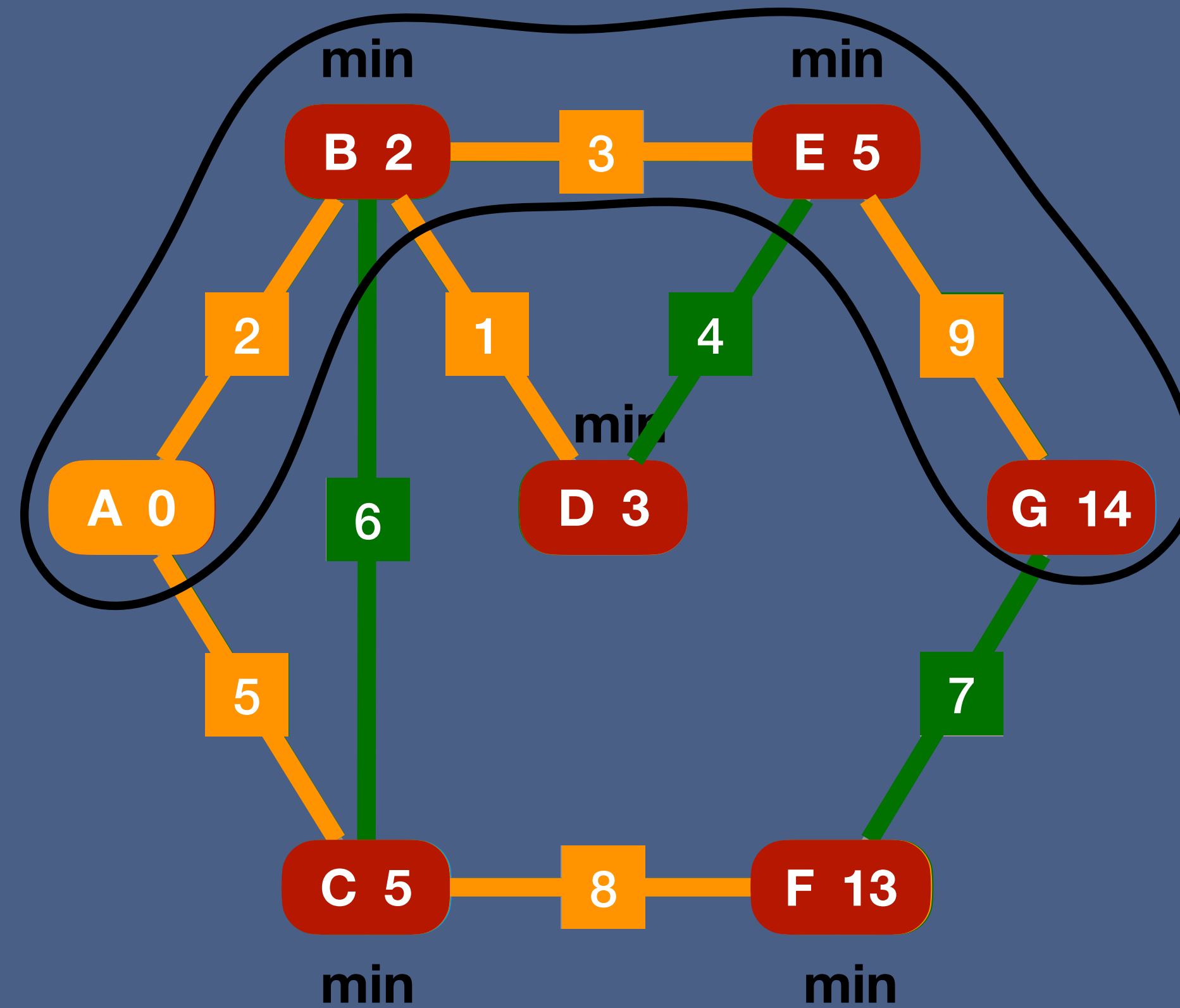


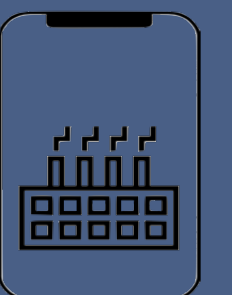
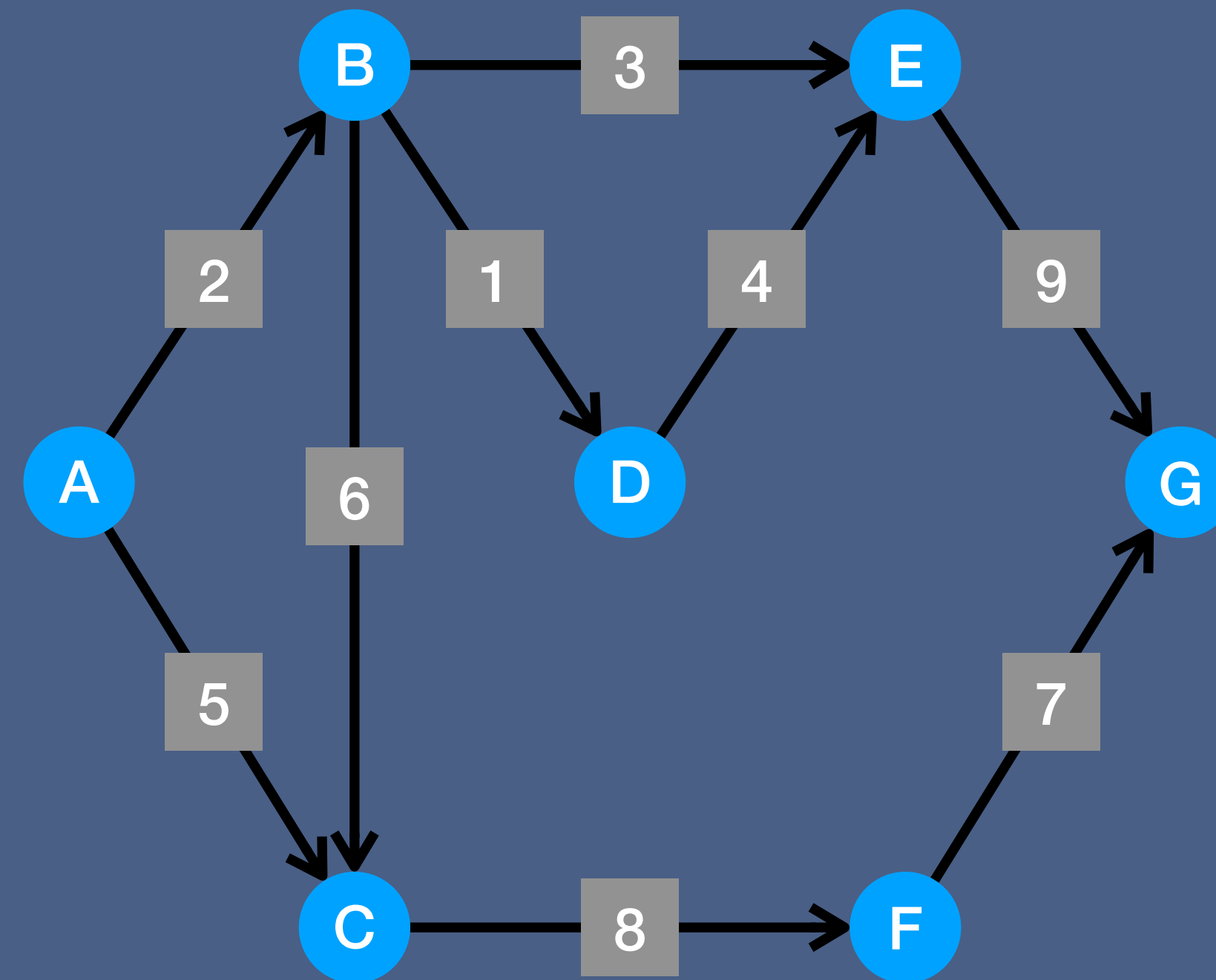
Dijkstra's Algorithm



Dijkstra's Algorithm for SSSPP



Dijkstra's Algorithm for SSSPP

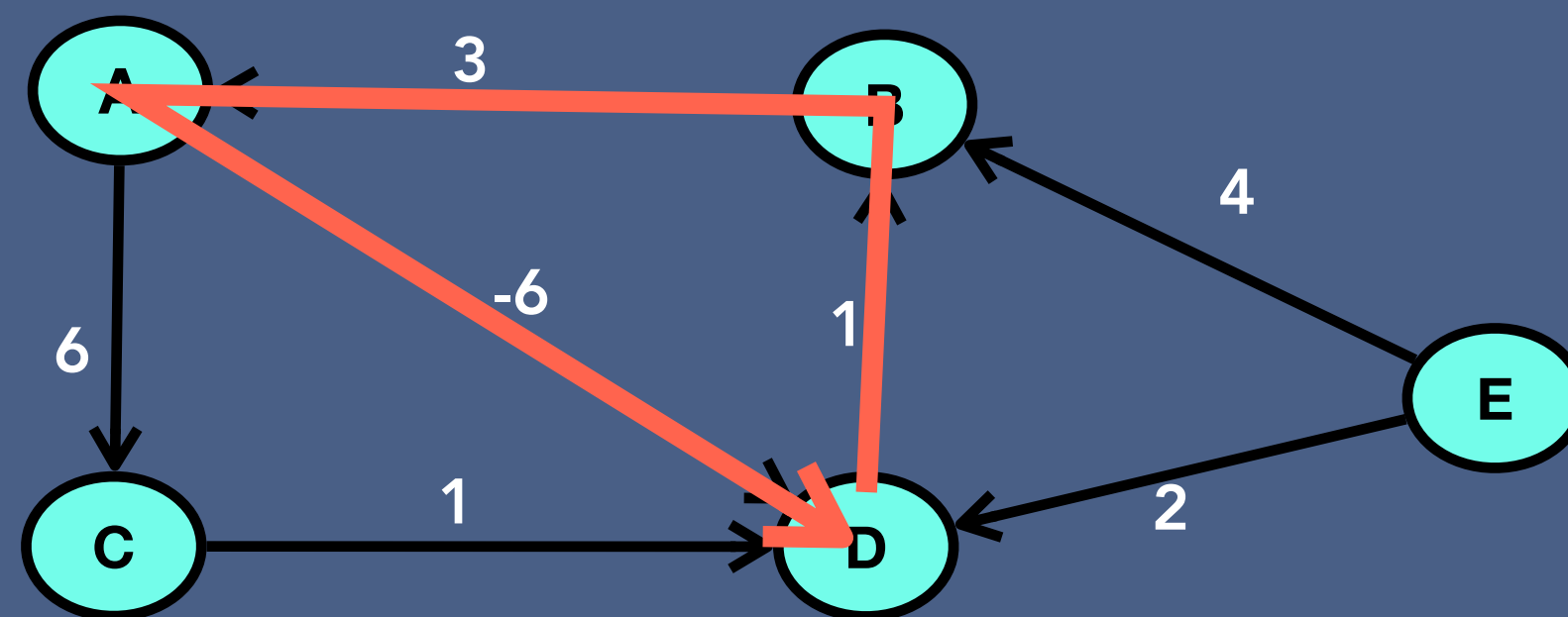


Dijkstra's Algorithm with Negative Cycle

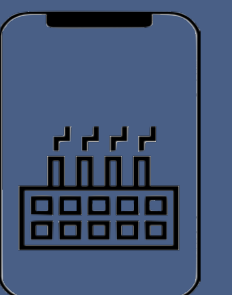
A path is called a negative cycle if:

There is a cycle (a cycle is a path of edges or vertices wherein a vertex is reachable from itself)

Total weight of cycle is a negative number

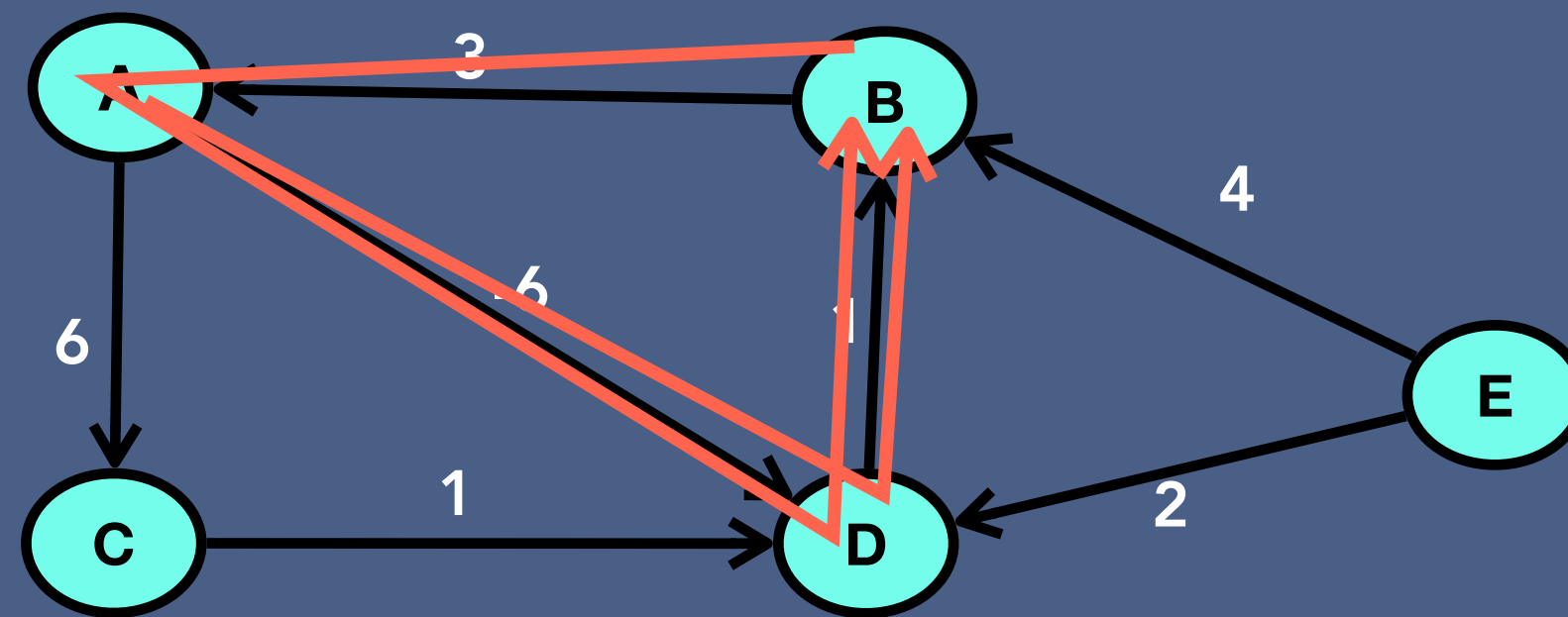


$$1 + 3 + (-6) = -2$$



Dijkstra's Algorithm with Negative Cycle

We cannot never find a negative cycle in a graph



Path from A to B = $6 + 1 = -5$
 $= -5 + 3 + (-6) + 1 = -7$
 $= -7 + 3 + (-6) + 1 = -9$
 $= -9 + 3 + (-6) + 1 = -11$

